

Unit 1.2 Ratio and Proportion

Paper 1: Basic

1. y is inversely proportional to x^2 .

Which of these diagrams represents the graph of y against x ?

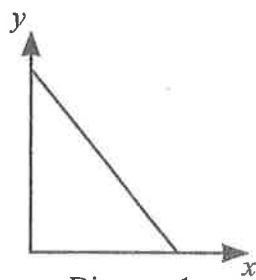


Diagram 1

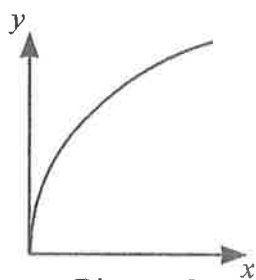


Diagram 2

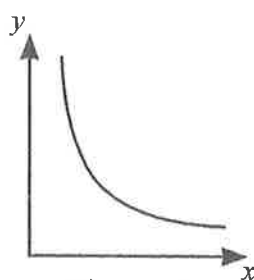


Diagram 3

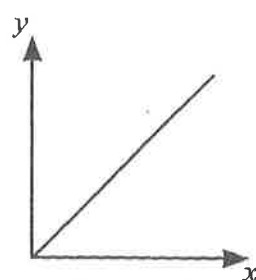


Diagram 4

[1]

[N20/I/2]

2. Kim, Pat and Xin share \$285.

The ratio of the amount that Kim receives to the **total** amount that Pat and Xin receive is 3 : 2.

Pat receives \$24 more than Xin.

Calculate how much they each receive.

[3]

[N20/I/12]

3. A map has a scale of 1 : 2 000 000.

- (a) The distance between Singapore and Phuket is 950 km.

Calculate the distance, in centimetres, between Singapore and Phuket on the map.

[2]

- (b) The area of Malaysia is 330 803 km².

Calculate the area, in square centimetres, of Malaysia on the map.

[2]

[N20/I/17]

4. A map of Western Europe has a scale of 1 : 2 500 000.

- (a) The length of the river Rhine on the map is 49.3 cm.

Calculate the actual length, in kilometres, of the river Rhine.

[2]

- (b) The area of Switzerland is 41 285 km².

Calculate the area, in square centimetres, of Switzerland on the map.

[2]

[N16/I/20]

5. (a) Wong makes a fruit drink.

He uses apple juice, peach juice and lemonade in the ratio 5 : 2 : 8 respectively.

He uses 2.4 litres of lemonade.

- (i) How much apple juice does he use?

[1]

- (ii) How much fruit drink does he make altogether?

[1]

- (b) Min makes a fruit drink using orange juice, lemonade and pineapple juice.

The ratio orange juice : lemonade is 2 : 5.

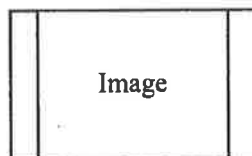
The ratio lemonade : pineapple juice is 3 : 4.

Find the ratio orange juice : lemonade : pineapple juice.

[1]

[N15/I/12]

6. A television screen has sides in the ratio width : height = 4 : 3.
 A laptop screen has sides in the ratio width : height = 8 : 5.
 An image exactly fills the television screen.
 The same image only fills the height of the laptop screen and not the width, as shown in the diagram.



Find the fraction of the laptop screen not covered by the image.
 Give your answer in its lowest terms.

[3]
 [N22/I/12]

7. The kinetic energy, E joules, of a moving object is directly proportional to the square of its speed, v m/s.
 The kinetic energy of a particular object is reduced by 75% of its original value.
 Calculate the percentage reduction in the speed of this object.

[2]
 [N21/I/16(b)]

8. P is directly proportional to Q^3 .

(a) When $Q = \frac{1}{3}$, $P = 2$.

Find P when $Q = \frac{1}{6}$.

[2]

(b) A change in Q produces a change in P .

When the value of Q is multiplied by $\frac{1}{2}$, the value of P changes by a factor of m .

Find m .

[1]
 [N19/I/10]

9. Ann and Ben each have a savings account.

The ratio Ann's savings : Ben's savings = 7 : 5.

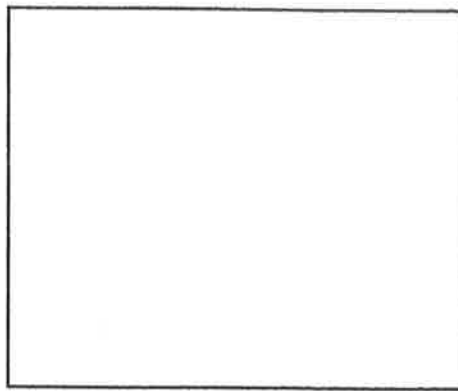
They each spend \$60 from their savings.

The new ratio Ann's savings : Ben's savings = 3 : 2.

Find how much money Ann now has in her account.

[4]
 [N19/I/19]

10.



Scale: 1 cm represents n metres

The diagram shows the floor plan of Cheryl's rectangular bedroom.

It is drawn to a scale of 1 cm represents n metres.

The actual area of the floor is 19.2 m^2 .

(a) Using the plan, find the value of n .

[2]

(b) Cheryl decides to varnish the floor in her bedroom.

The varnish she needs is sold in tins of three sizes.

250 ml	500 ml	1 litre
\$24.00	\$42.50	\$85.95

1 litre of varnish covers 16 m^2 .

Calculate the least amount of money that Cheryl needs to pay for the varnish.

[3]

[N19/I/23]

11. (a) The time taken to erect a fence is inversely proportional to the number of workers.

Four workers take 17.5 hours.

Find the time it would have taken 5 workers to erect the fence.

[1]

(b) The force, F , between two objects is inversely proportional to the square of the distance, d , between them.

The distance between two objects is increased by 20%.

Calculate the percentage reduction in the force between the objects.

[2]

[N18/I/12]

12.



A company manufactures three sizes of the same brand of shampoo.

Show that the cost of the shampoo is **not** directly proportional to the quantity of shampoo.

[2]

[N17/I/23(a)]

